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RESEARCH AND TECHNOLOGY INSTITUTE  
LABORATORY OF DATA ENGINEERING

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# Web Application for DPCfam and DPCstruct Data Exploration

Report 3 (27 Jan - 2 Feb 2026) : Web Development  
(DPCFam Backend)

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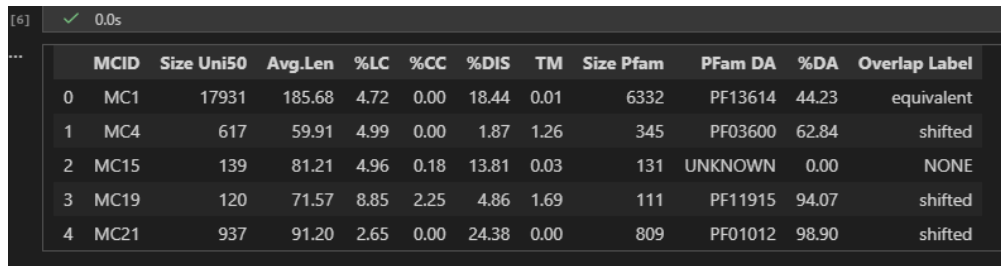
# 1 Introduction

This report is the third in a sequence describing the completion of weekly assigned tasks during our internship. Focusing on the DPCFam dataset(Russo et al., 2022), we present the backend methodology used to produce the first draft of the DPCFam Metaclusters web page, where users can explore properties related to DPCFam MCs. A local PostgreSQL database is used for development purposes.

## 2 Development

### 2.1 Database Creation

First, we developed a series of notebooks(available [here](#)) aimed at converting each .txt file in the dpcfam/ directory into a .csv format. These files were subsequently merged into a single dataframe whose columns represent relevant metacluster properties, as shown in Figure 1:



The screenshot shows a Jupyter Notebook interface with a table of DPCFam Metaclusters properties. The table has 12 columns: MCID, Size Uni50, Avg.Len, %LC, %CC, %DIS, TM, Size Pfam, PFam DA, %DA, Overlap, and Label. The data is as follows:

|   | MCID | Size Uni50 | Avg.Len | %LC  | %CC  | %DIS  | TM   | Size Pfam | PFam DA | %DA   | Overlap Label |
|---|------|------------|---------|------|------|-------|------|-----------|---------|-------|---------------|
| 0 | MC1  | 17931      | 185.68  | 4.72 | 0.00 | 18.44 | 0.01 | 6332      | PF13614 | 44.23 | equivalent    |
| 1 | MC4  | 617        | 59.91   | 4.99 | 0.00 | 1.87  | 1.26 | 345       | PF03600 | 62.84 | shifted       |
| 2 | MC15 | 139        | 81.21   | 4.96 | 0.18 | 13.81 | 0.03 | 131       | UNKNOWN | 0.00  | NONE          |
| 3 | MC19 | 120        | 71.57   | 8.85 | 2.25 | 4.86  | 1.69 | 111       | PF11915 | 94.07 | shifted       |
| 4 | MC21 | 937        | 91.20   | 2.65 | 0.00 | 24.38 | 0.00 | 809       | PF01012 | 98.90 | shifted       |

Figure 1: DPCFam Metaclusters properties. We noticed that the Size Pfam column doesn't match Federico's web page, this will be revisited.

We then installed a PostgreSQL server on the local machine :

```
=====
# Installation (WSL Ubuntu)
sudo apt update
sudo apt install postgresql postgresql-contrib
# Check version :
psql --version
# Status / Start / Stop
sudo service postgresql status
sudo service postgresql start
sudo -u postgres psql # switch to postgres user
# sudo service postgresql stop
# \l (list databases), \c A (connect to database A),
# \dt (list tables), \q (quit)
=====
```

We created a database (`dpcfam_mcs_db`), a user(`enyanduk`) and a table (`mcs_properties`):

1. Create a database and user :

```
=====
CREATE DATABASE dpcfam_mcs_db;
CREATE USER enyanduk WITH PASSWORD 'EmmaPSQL2026';
ALTER ROLE enyanduk SET client_encoding TO 'utf8';
ALTER ROLE enyanduk SET default_transaction_isolation TO 'read committed';
ALTER ROLE enyanduk SET timezone TO 'UTC';
GRANT ALL PRIVILEGES ON DATABASE mydatabase TO enyanduk;
=====
```

2. Create table :

Table 1: MCs properties table in PostgreSQL

```
=====
CREATE TABLE mcs_properties (
    mcid TEXT PRIMARY KEY,
    size_uniref50 INTEGER,
    avg_len DOUBLE PRECISION,
    lc_percent DOUBLE PRECISION,
    cc_percent DOUBLE PRECISION,
    dis_percent DOUBLE PRECISION,
    tm DOUBLE PRECISION,
    size_pfam INTEGER,
    pfam_da TEXT,
    da_percent DOUBLE PRECISION,
    overlap_label TEXT
);
=====
```

3. Copy data from our csv into the table by skipping the header:

```
=====
\copy mcs_properties
FROM '/tmp/dpcfam_mcs_props.csv'
CSV HEADER;
=====
```

## 2.2 DPCFam Metaclusters Web Page

Clone the repository, move into it, activate the virtual environment, and install dependencies:

```
git clone https://github.com/emmanuelnyandukagarabi/dpc_fam_and_struct_webapp
cd dpc_fam_and_struct_webapp && source .area_internship_venv/bin/activate
pip install -r requirements.txt
```

### 2.2.1 Installation

To allow Django to connect to PostgreSQL, the **psycopg2-binary** library was installed in the virtual environment, and we updated the database configuration:

(/dpc\_fam\_and\_struct\_webapp/settings.py)

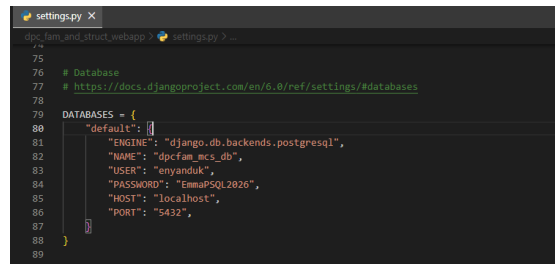


Figure 2: Database settings. PostgreSQL is used instead of the default SQLite3 backend.

### 2.2.2 Model : dpcfam/models.py

The Model represents our PostgreSQL table (Table 1) in Django:

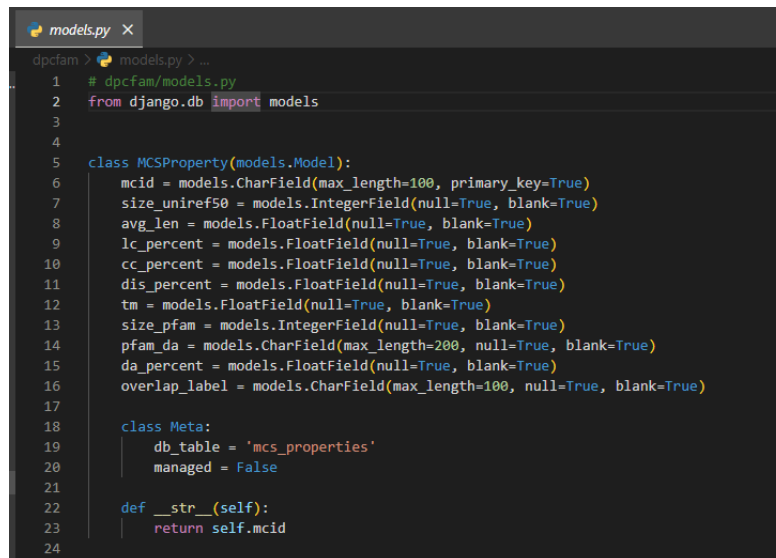
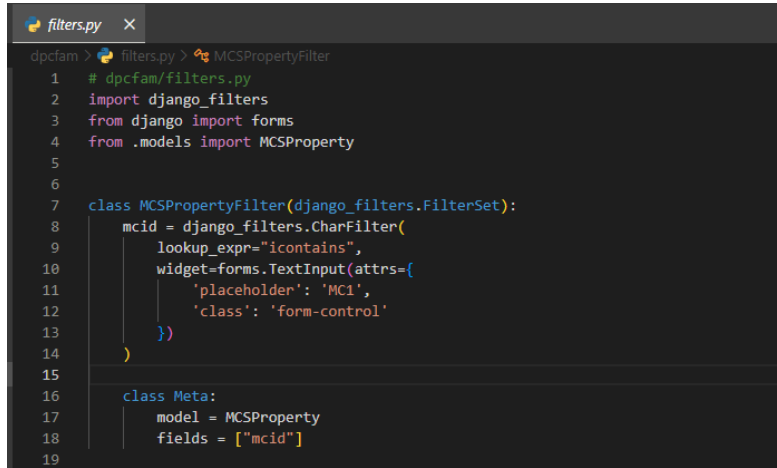


Figure 3: DPCFam MCSPROPERTY Model : Each field in the model corresponds to a column in our PostgreSQL table (Table 1), we do not allow django to create/modify/delete it via migrations since it already exists(`managed=False`), MCID is used as the primary key (no need to autoincrement `id`), and when django displays an instance, it shows the MCID.

### 2.2.3 Filter : dpcfam/filters.py

Filtering enables users to search for specific metaclusters:

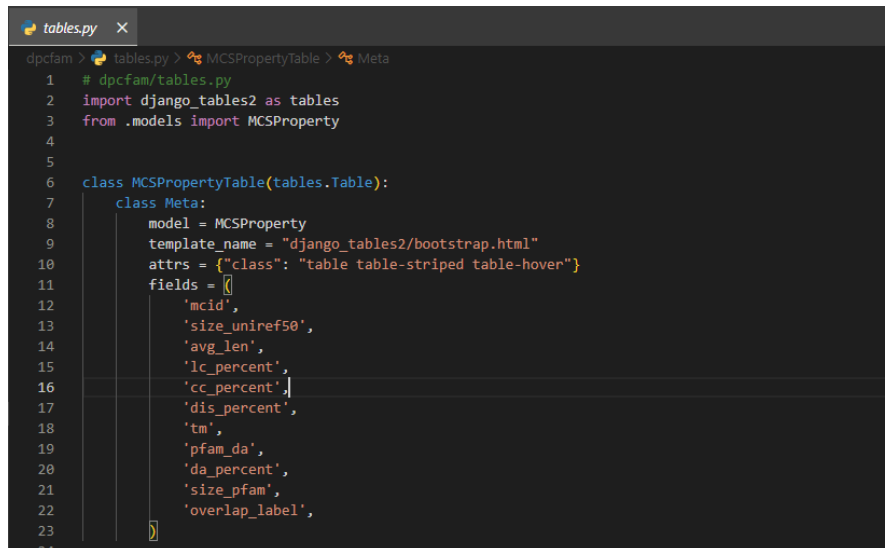


```
filters.py X
dpcfam > filters.py > MCSPropertyFilter
1 # dpcfam/filters.py
2 import django_filters
3 from django import forms
4 from .models import MCSProperty
5
6
7 class MCSPropertyFilter(django_filters.FilterSet):
8     mcid = django_filters.CharFilter(
9         lookup_expr="icontains",
10         widget=forms.TextInput(attrs={
11             'placeholder': 'MC1',
12             'class': 'form-control'
13         })
14     )
15
16 class Meta:
17     model = MCSProperty
18     fields = ["mcid"]
19
```

Figure 4: DPCFam Filters : A user enters a MCID, we display all MCIDs which contain ID, a form is then submitted to call a view with ?mcid=MCID in the URL.

### 2.2.4 Tables : dpcfam/tables.py

This module defines which fields are displayed and how they are rendered.

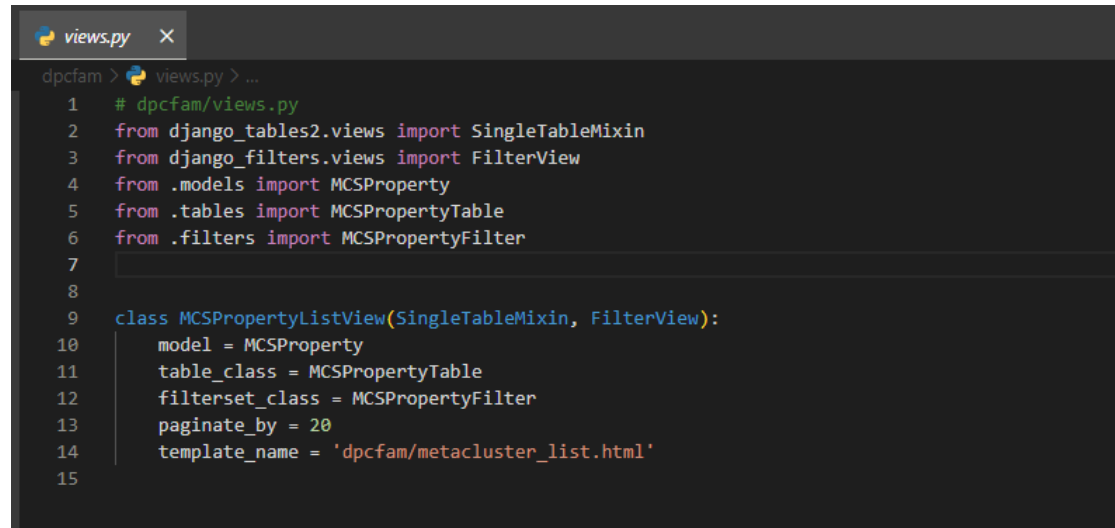


```
tables.py X
dpcfam > tables.py > MCSPropertyTable > Meta
1 # dpcfam/tables.py
2 import django_tables2 as tables
3 from .models import MCSProperty
4
5
6 class MCSPropertyTable(tables.Table):
7     class Meta:
8         model = MCSProperty
9         template_name = "django_tables2/bootstrap.html"
10        attrs = {"class": "table table-striped table-hover"}
11        fields = [
12            'mcid',
13            'size_uniref50',
14            'avg_len',
15            'lc_percent',
16            'cc_percent',
17            'dis_percent',
18            'tm',
19            'pfam_da',
20            'da_percent',
21            'size_pfam',
22            'overlap_label',
23        ]
24
```

Figure 5: DPCFam Tables : We specify which model fields to display as table columns, we then use django\_tables2 library to autogenerate an HTML table which is styled using Bootstrap template.

### 2.2.5 Views : dpcfam/views.py

Views integrate data retrieval, filtering, pagination, and rendering:

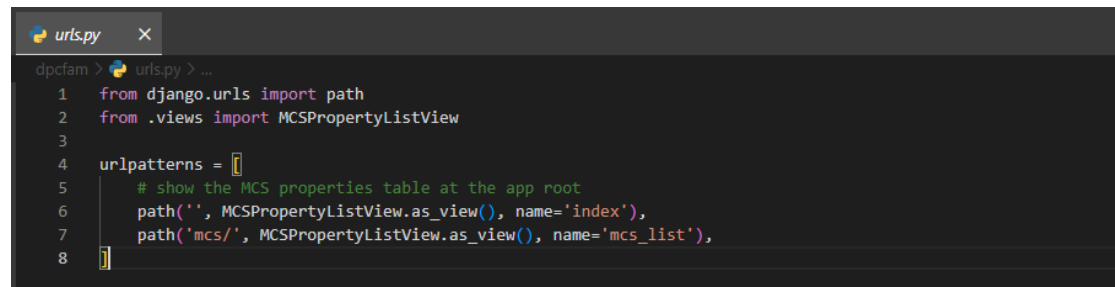


```
views.py x
dpcfam > views.py > ...
1 # dpcfam/views.py
2 from django_tables2.views import SingleTableMixin
3 from django_filters.views import FilterView
4 from .models import MCSProperty
5 from .tables import MCSPropertyTable
6 from .filters import MCSPropertyFilter
7
8
9 class MCSPropertyListView(SingleTableMixin, FilterView):
10     model = MCSProperty
11     table_class = MCSPropertyTable
12     filterset_class = MCSPropertyFilter
13     paginate_by = 20
14     template_name = 'dpcfam/metacluster_list.html'
15
```

Figure 6: DPCFam Views : The rendered HTML table (MCSPropertyTable) and filter form (MCSPropertyFilter) are combined and paginated (20 rows per page), then passed to the template.

### 2.2.6 URLs : dpcfam/urls.py

URLs route HTTP requests to appropriate views:



```
urls.py x
dpcfam > urls.py > ...
1 from django.urls import path
2 from .views import MCSPropertyListView
3
4 urlpatterns = []
5 # show the MCS properties table at the app root
6 path('', MCSPropertyListView.as_view(), name='index'),
7 path('mcs/', MCSPropertyListView.as_view(), name='mcs_list'),
8
```

Figure 7: DPCFam URLs : When a user visits "/" or "/mcs/", Django calls MCSPropertyListView, processes the GET request, and renders the template.

### 2.2.7 Template : dpcfam/templates/dpcfam/metacluster\_list.html

The HTML template renders the page that the user sees in their browser:

```

dj metacluster_list.html x
dpcfam > templates > dpcfam > dj metacluster_list.html
22 </head>
23 <body>
24 <div class="container py-4">
25 <h1 class="mb-3">Metaclusters</h1>
26
27 <form method="get" class="mb-3">
28 <div class="d-flex align-items-center gap-2">
29 <label for="id_mcid" class="mb-0 flex-shrink-0">Enter MCID</label>
30 <div style="flex: 0 0 auto;">
31 <{{ filter.form.mcid }}>
32 </div>
33 <button type="submit" class="btn btn-primary flex-shrink-0">Search</button>
34 </div>
35 </form>
36
37 <{% render_table table %}>
38
39 <!-- Pagination Controls -->

```

Figure 8: DPCFam template : Basically, the filter form submits query parameters via the URL (?mcid=<value>), and Django renders a paginated table with proper structure and styling.

### 2.2.8 Admin : dpcfam/admin.py

The admin interface allows staff users to manage data:

```

admin.py x
dpcfam > admin.py x MCSPropertyAdmin
1 from django.contrib import admin
2 from .models import MCSProperty
3
4
5 @admin.register(MCSProperty)
6 class MCSPropertyAdmin(admin.ModelAdmin):
7     list_display = (
8         'mcid',
9         'size_uniref50',
10        'pfam_da',
11        'overlap_label',
12    )
13    search_fields = ('mcid', 'pfam_da')
14

```

(a) Admin Registry

| MCID    | Size Uniref50 | Pfam DA | Overlap Label |
|---------|---------------|---------|---------------|
| MC20000 | 115           | PF13025 | unifam        |
| MC20001 | 54            | PF13024 | extended      |
| MC20002 | 102           | PF13023 | extended      |

(b) Admin interface

Figure 9: DPCFam admin interface : Staff users (created using `python3 manage.py createsuperuser`) can log in at `"/admin/"` to view, filter, search (by MCID or Pfam DA), and edit metacluster properties.

## 2.2.9 Final Result: Paginated DPCFam Metaclusters Table

127.0.0.1:8000/dpcfam/

### Metaclusters

Enter MCID  Search

| McId  | Size_uniref50 | Avg_len | Lc_percent | Cc_percent | Dis_percent | Tm   | Pfam_da | Da_percent | Size_pfam | Overlap_label |
|-------|---------------|---------|------------|------------|-------------|------|---------|------------|-----------|---------------|
| MC1   | 17931         | 185.68  | 4.72       | 0.0        | 18.44       | 0.01 | PF13614 | 44.23      | 6332      | equivalent    |
| MC4   | 617           | 59.91   | 4.99       | 0.0        | 1.87        | 1.26 | PF03600 | 62.84      | 345       | shifted       |
| MC15  | 139           | 81.21   | 4.96       | 0.18       | 13.81       | 0.03 | UNKNOWN | 0.0        | 131       | NONE          |
| MC19  | 120           | 71.57   | 8.85       | 2.25       | 4.86        | 1.69 | PF11915 | 94.07      | 111       | shifted       |
| MC21  | 937           | 91.2    | 2.65       | 0.0        | 24.38       | 0.0  | PF01012 | 98.9       | 809       | shifted       |
| MC22  | 51            | 173.59  | 0.92       | 0.0        | 24.13       | 0.07 | PF13391 | 10.81      | 4         | shifted       |
| MC24  | 77            | 89.6    | 6.27       | 0.0        | 0.43        | 2.09 | PF12698 | 6.56       | 4         | shifted       |
| MC27  | 132           | 58.77   | 0.43       | 0.0        | 7.79        | 0.0  | PF00884 | 10.85      | 14        | shifted       |
| MC31  | 252           | 280.19  | 3.94       | 0.0        | 24.36       | 0.08 | PF12937 | 0.4        | 1         | shifted       |
| MC58  | 98            | 64.79   | 18.28      | 0.0        | 3.91        | 0.87 | PF00953 | 36.36      | 28        | shifted       |
| MC64  | 52            | 88.85   | 0.78       | 0.0        | 35.45       | 0.0  | PF18911 | 2.63       | 1         | shifted       |
| MC67  | 121           | 180.2   | 4.02       | 0.0        | 14.81       | 0.92 | PF09451 | 2.13       | 2         | shifted       |
| MC73  | 76            | 153.57  | 0.78       | 0.0        | 14.82       | 0.0  | UNKNOWN | 0.0        | 61        | NONE          |
| MC77  | 118           | 60.57   | 1.57       | 0.0        | 34.78       | 0.0  | UNKNOWN | 0.0        | 73        | NONE          |
| MC82  | 59            | 191.24  | 6.79       | 0.0        | 6.72        | 0.0  | UNKNOWN | 0.0        | 59        | NONE          |
| MC84  | 89            | 69.96   | 0.16       | 0.0        | 7.57        | 0.0  | PF13579 | 4.41       | 3         | shifted       |
| MC95  | 184           | 68.38   | 1.06       | 0.0        | 12.97       | 0.02 | PF02705 | 4.73       | 7         | shifted       |
| MC96  | 77            | 128.54  | 4.64       | 0.0        | 21.67       | 0.03 | PF04434 | 18.18      | 14        | extended      |
| MC100 | 81            | 107.57  | 0.62       | 0.0        | 17.4        | 0.0  | UNKNOWN | 0.0        | 61        | NONE          |
| MC103 | 107           | 121.42  | 2.2        | 0.0        | 27.3        | 0.0  | UNKNOWN | 0.0        | 99        | NONE          |

12345678...2342next >

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Figure 10: DPCFam Metaclusters page : After running `python3 manage.py (makemigrations , migrate , runserver)`, we may visit the page at <http://127.0.0.1:8000/>.

## 3 Conclusion

To display this draft version, a local PostgreSQL database configured as described above is required. We will shortly extend the application by introducing additional tables (MCID, Protein, Length, Amino Acids) and establishing relationships between them.

## References

Russo, E. T., Barone, F., Bateman, A., Cozzini, S., Punta, M., and Laio, A. (2022). Dpcfam: unsupervised protein family classification by density peak clustering of large sequence datasets. *PLOS Computational Biology*, 18(10):e1010610.